# *Implementing CVPR16 “Deep hierarchical temporal model for group recognition”*

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## 1. Dataset

- The data consists of 55 video clips of volleyball matches from the olympiad, since it is high quality videos
- Dataset has 2 levels of annotation (9 person classes, 8 image classes)
- Dataset maybe has limitation in variety of characteristics of uniforms of players
- Around of 80% of the data for Training and 20% for testing(2102 : 459)
- Training test split indices is random every run

| Action Classes | No. of Instances |
|---|---|
| Waiting | 3601 |
| Setting | 1332 |
| Digging | 2333 |
| Falling | 1241 |
| Spiking | 1216 |
| Blocking | 2458 |
| Jumping | 341 |
| Moving | 5121 |
| Standing | 38696 |

| Group Activity Class | No. of Instances |
|---|---|
| Right set | 644 |
| Right spike | 623 |
| Right pass | 801 |
| Right winpoint | 295 |
| Left winpoint | 367 |
| Left pass | 826 |
| Left spike | 642 |
| Left set | 633 |

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## 2. Experiments

- **First exp(baseline1 : image classifier); I used (learning rate = 1e-4, Epochs = 20, batch_size = 32) | training videos are** (12, 24, 34, 33, 28, 40, 25, 27, 6, 44, 23, 21, 13, 31, 10, 7, 1, 9, 37, 41, 49, 15, 53, 46, 18, 38, 32, 4, 29, 54, 52, 12, 22, 0, 39, 2, 5, 51, 45, 20, 16, 35, 3, 14 | **testing videos are** (42, 48, 36, 26, 47, 11, 50, 30, 8, 19, 43) | **Achieving 80.8% on testset**

```
_ Starting Baseline B1 Image Model Training...
Epoch [1/20] -> Loss: 1.8118 | Training Accuracy: 23.88%
Epoch [2/20] -> Loss: 1.5199 | Training Accuracy: 28.64%
Epoch [3/20] -> Loss: 1.1199 | Training Accuracy: 50.19%
Epoch [4/20] -> Loss: 0.8437 | Training Accuracy: 59.32%
Epoch [5/20] -> Loss: 0.6504 | Training Accuracy: 72.41%
Epoch [6/20] -> Loss: 0.4696 | Training Accuracy: 78.97%
Epoch [7/20] -> Loss: 0.3705 | Training Accuracy: 83.30%
Epoch [8/20] -> Loss: 0.3237 | Training Accuracy: 85.82%
Epoch [9/20] -> Loss: 0.2283 | Training Accuracy: 90.77%
Epoch [10/20] -> Loss: 0.1545 | Training Accuracy: 94.62%
Epoch [11/20] -> Loss: 0.1567 | Training Accuracy: 95.05%
Epoch [12/20] -> Loss: 0.1003 | Training Accuracy: 96.67%
Epoch [13/20] -> Loss: 0.1344 | Training Accuracy: 95.91%
Epoch [14/20] -> Loss: 0.1107 | Training Accuracy: 96.81%
Epoch [15/20] -> Loss: 0.0593 | Training Accuracy: 98.43%
Epoch [16/20] -> Loss: 0.0513 | Training Accuracy: 98.38%
Epoch [17/20] -> Loss: 0.0535 | Training Accuracy: 98.38%
Epoch [18/20] -> Loss: 0.0397 | Training Accuracy: 98.95%
Epoch [19/20] -> Loss: 0.0425 | Training Accuracy: 98.53%
Epoch [20/20] -> Loss: 0.0393 | Training Accuracy: 98.95%

🔬 Evaluating Model Performance Against Unseen Test Set...
📊 Final Accuracy on the Test Set: 80.83%
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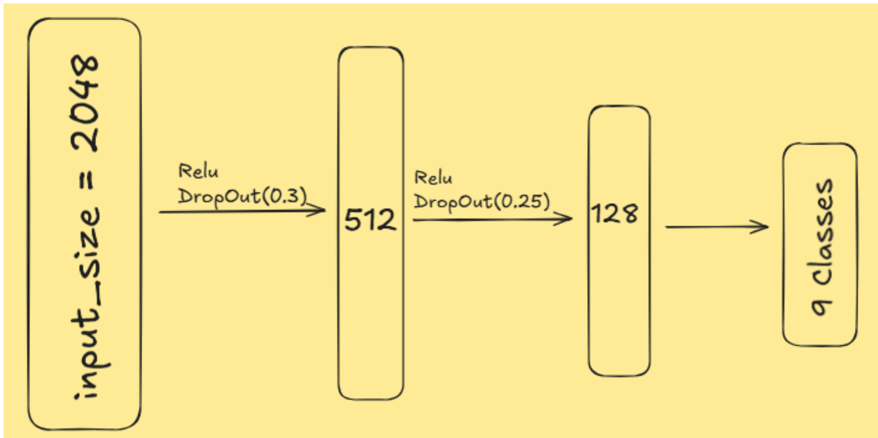
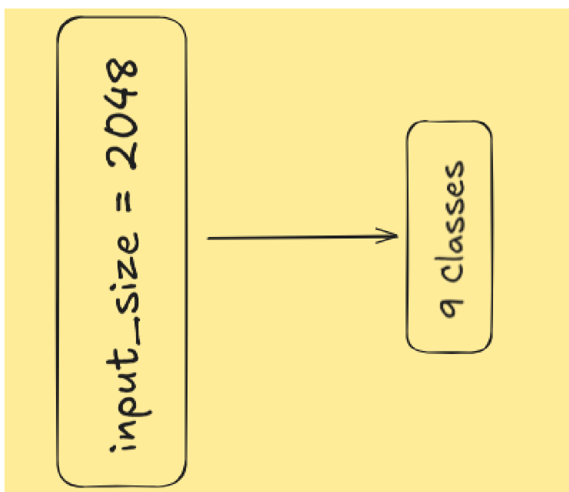
**<u>Notes:</u>** same model without fine-tuning the backbone(resnet50), just change the classifier layers achieved 53% - 60% (depending on the clips and learning rate)

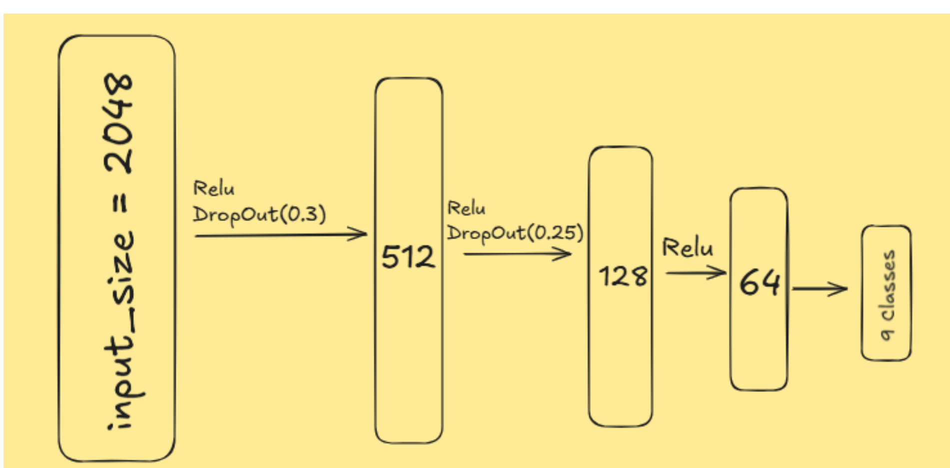
- **Second exp(baseline 2: Person classifier); I used (Learning rate= 0.001, Epochs = 10, batch_size = 32)** Feed each person to classifier into 9 classes(person level)
  - Classifier one: **achieved 76.63% on test set**




  - Classifier two: **achieved 74.23% on test set**




  - Classifier three: **achieved 77.23% on test set**



```
Starting Baseline B2 Model Fine-Tuning...
Epoch 01/10 | Loss: 0.9239 | Train Acc: 72.51%
Epoch 02/10 | Loss: 0.8465 | Train Acc: 74.25%
Epoch 03/10 | Loss: 0.8168 | Train Acc: 74.94%
Epoch 04/10 | Loss: 0.7946 | Train Acc: 75.44%
Epoch 05/10 | Loss: 0.7773 | Train Acc: 75.85%
Epoch 06/10 | Loss: 0.7626 | Train Acc: 76.18%
Epoch 07/10 | Loss: 0.7492 | Train Acc: 76.51%
Epoch 08/10 | Loss: 0.7374 | Train Acc: 76.82%
Epoch 09/10 | Loss: 0.7269 | Train Acc: 76.99%
Epoch 10/10 | Loss: 0.7182 | Train Acc: 77.24%

Evaluating Model Performance Against Unseen Test Set...
Final Accuracy on the Test Set: 77.23%
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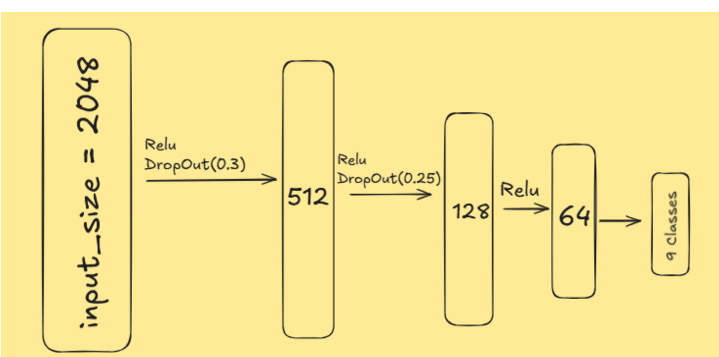
- **third exp(**baseline 3: Image classifier on Pooled people); **I used( learning rate =0.001, batch_size = 64, Epochs = 15)** ; The output sequence from feature-extraction file is tensor(9, 12, 2048), 9 frames, 12 people, 2048 features.

**So Firstly**; Avg pooling through frames -> tensor(12, 2048).

**Secondly**; Max pooling through players -> tensor(1, 2048): good representation for whole image

**Lastly**; feed the (1, 2048) to image classifier on 8 group activity classes.

- Classifier one:




```
Epoch 01/15 | Loss: 2.0132 | Train Acc: 18.77%
Epoch 02/15 | Loss: 1.8289 | Train Acc: 26.09%
Epoch 03/15 | Loss: 1.5786 | Train Acc: 36.11%
Epoch 04/15 | Loss: 1.4739 | Train Acc: 39.83%
Epoch 05/15 | Loss: 1.4026 | Train Acc: 43.04%
Epoch 06/15 | Loss: 1.3541 | Train Acc: 44.67%
Epoch 07/15 | Loss: 1.3117 | Train Acc: 47.82%
Epoch 08/15 | Loss: 1.2721 | Train Acc: 50.27%
Epoch 09/15 | Loss: 1.2112 | Train Acc: 52.27%
Epoch 10/15 | Loss: 1.1635 | Train Acc: 53.48%
Epoch 11/15 | Loss: 1.0862 | Train Acc: 58.87%
Epoch 12/15 | Loss: 1.0618 | Train Acc: 61.35%
Epoch 13/15 | Loss: 1.0116 | Train Acc: 62.65%
Epoch 14/15 | Loss: 1.0364 | Train Acc: 60.56%
Epoch 15/15 | Loss: 0.9804 | Train Acc: 64.13%

Evaluating Model Performance Against Unseen Test Set...
Final Accuracy on the Test Set: 50.42%
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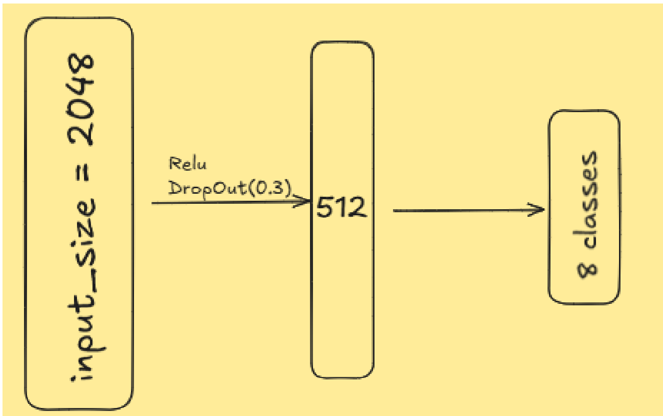
- Classifier two




```
Epoch 01/15 | Loss: 1.8674 | Train Acc: 27.03%
Epoch 02/15 | Loss: 1.5675 | Train Acc: 40.10%
Epoch 03/15 | Loss: 1.4190 | Train Acc: 47.28%
Epoch 04/15 | Loss: 1.3066 | Train Acc: 52.27%
Epoch 05/15 | Loss: 1.2129 | Train Acc: 55.63%
Epoch 06/15 | Loss: 1.1710 | Train Acc: 57.84%
Epoch 07/15 | Loss: 1.1204 | Train Acc: 60.68%
Epoch 08/15 | Loss: 1.0530 | Train Acc: 63.08%
Epoch 09/15 | Loss: 0.9939 | Train Acc: 66.01%
Epoch 10/15 | Loss: 0.9576 | Train Acc: 66.77%
Epoch 11/15 | Loss: 0.8973 | Train Acc: 69.16%
Epoch 12/15 | Loss: 0.8934 | Train Acc: 69.85%
Epoch 13/15 | Loss: 0.8334 | Train Acc: 71.76%
Epoch 14/15 | Loss: 0.7877 | Train Acc: 73.85%
Epoch 15/15 | Loss: 0.7532 | Train Acc: 74.27%

🔬 Evaluating Model Performance Against Unseen Test Set...
📊 Final Accuracy on the Test Set: 60.70%
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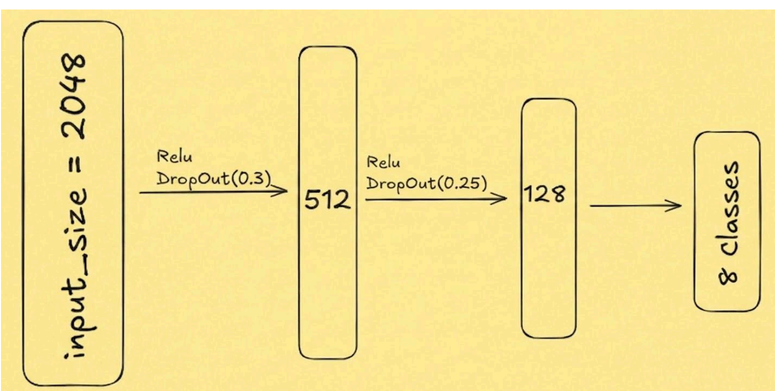
- Classifier three:




```
Epoch 01/15 | Loss: 1.9914 | Train Acc: 18.89%
Epoch 02/15 | Loss: 1.7683 | Train Acc: 29.57%
Epoch 03/15 | Loss: 1.5697 | Train Acc: 38.80%
Epoch 04/15 | Loss: 1.4800 | Train Acc: 41.59%
Epoch 05/15 | Loss: 1.3850 | Train Acc: 46.82%
Epoch 06/15 | Loss: 1.3176 | Train Acc: 48.82%
Epoch 07/15 | Loss: 1.2801 | Train Acc: 50.18%
Epoch 08/15 | Loss: 1.2120 | Train Acc: 55.08%
Epoch 09/15 | Loss: 1.1688 | Train Acc: 57.17%
Epoch 10/15 | Loss: 1.0922 | Train Acc: 58.57%
Epoch 11/15 | Loss: 1.0610 | Train Acc: 61.11%
Epoch 12/15 | Loss: 1.0038 | Train Acc: 64.47%
Epoch 13/15 | Loss: 0.9561 | Train Acc: 65.56%
Epoch 14/15 | Loss: 0.9411 | Train Acc: 65.80%
Epoch 15/15 | Loss: 0.9258 | Train Acc: 66.59%

🔬 Evaluating Model Performance Against Unseen Test Set...
📊 Final Accuracy on the Test Set: 55.02%
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- **Last and best Classifier:** new approach

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Epoch 01/15 | Loss: 1.4925 | Train Acc: 46.64%
Epoch 02/15 | Loss: 0.9072 | Train Acc: 71.67%
Epoch 03/15 | Loss: 0.5788 | Train Acc: 83.38%
Epoch 04/15 | Loss: 0.3616 | Train Acc: 90.25%
Epoch 05/15 | Loss: 0.2241 | Train Acc: 94.49%
Epoch 06/15 | Loss: 0.1281 | Train Acc: 97.19%
Epoch 07/15 | Loss: 0.0984 | Train Acc: 97.88%
Epoch 08/15 | Loss: 0.0870 | Train Acc: 97.91%
Epoch 09/15 | Loss: 0.0660 | Train Acc: 98.70%
Epoch 10/15 | Loss: 0.0652 | Train Acc: 98.31%
Epoch 11/15 | Loss: 0.0654 | Train Acc: 98.27%
Epoch 12/15 | Loss: 0.0616 | Train Acc: 98.37%
Epoch 13/15 | Loss: 0.0474 | Train Acc: 98.97%
Epoch 14/15 | Loss: 0.0460 | Train Acc: 98.64%
Epoch 15/15 | Loss: 0.0357 | Train Acc: 99.33%

🔬 Evaluating Model Performance Against Unseen Test Set...
📊 Final Accuracy on the Test Set: 71.10%
```
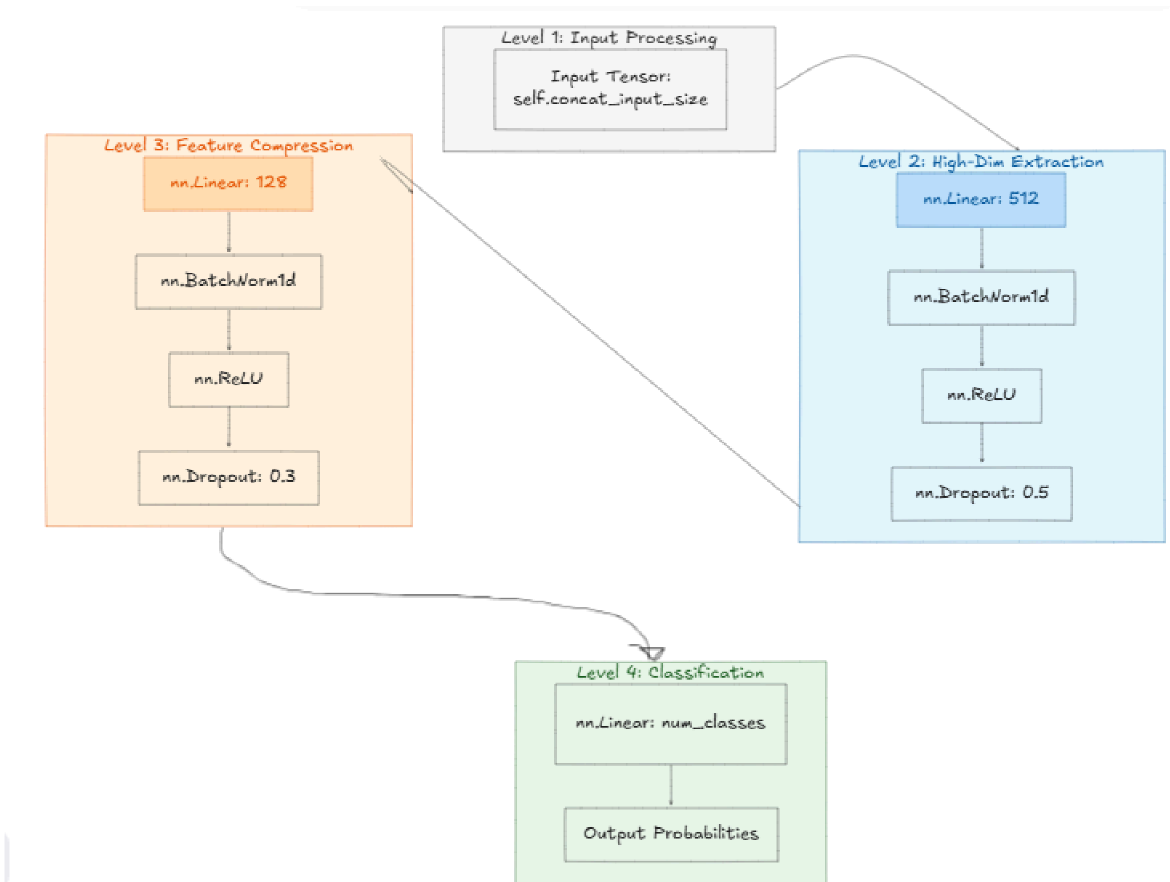

**NOTE**: there are some configuration i made make model achieve 72.8%; ( Added scheduler(step_size = 3, gamma = 0.5), weight_decay = 1e-3)

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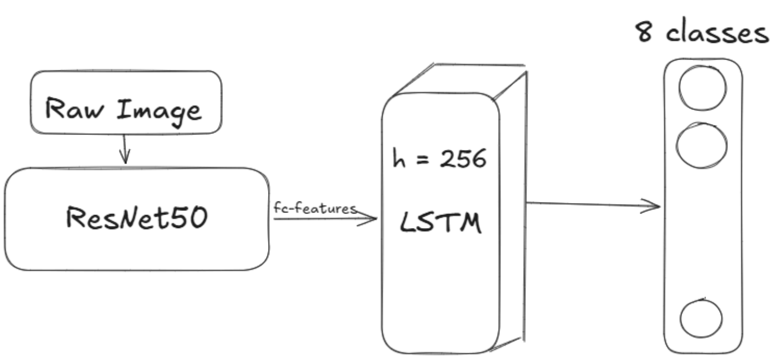
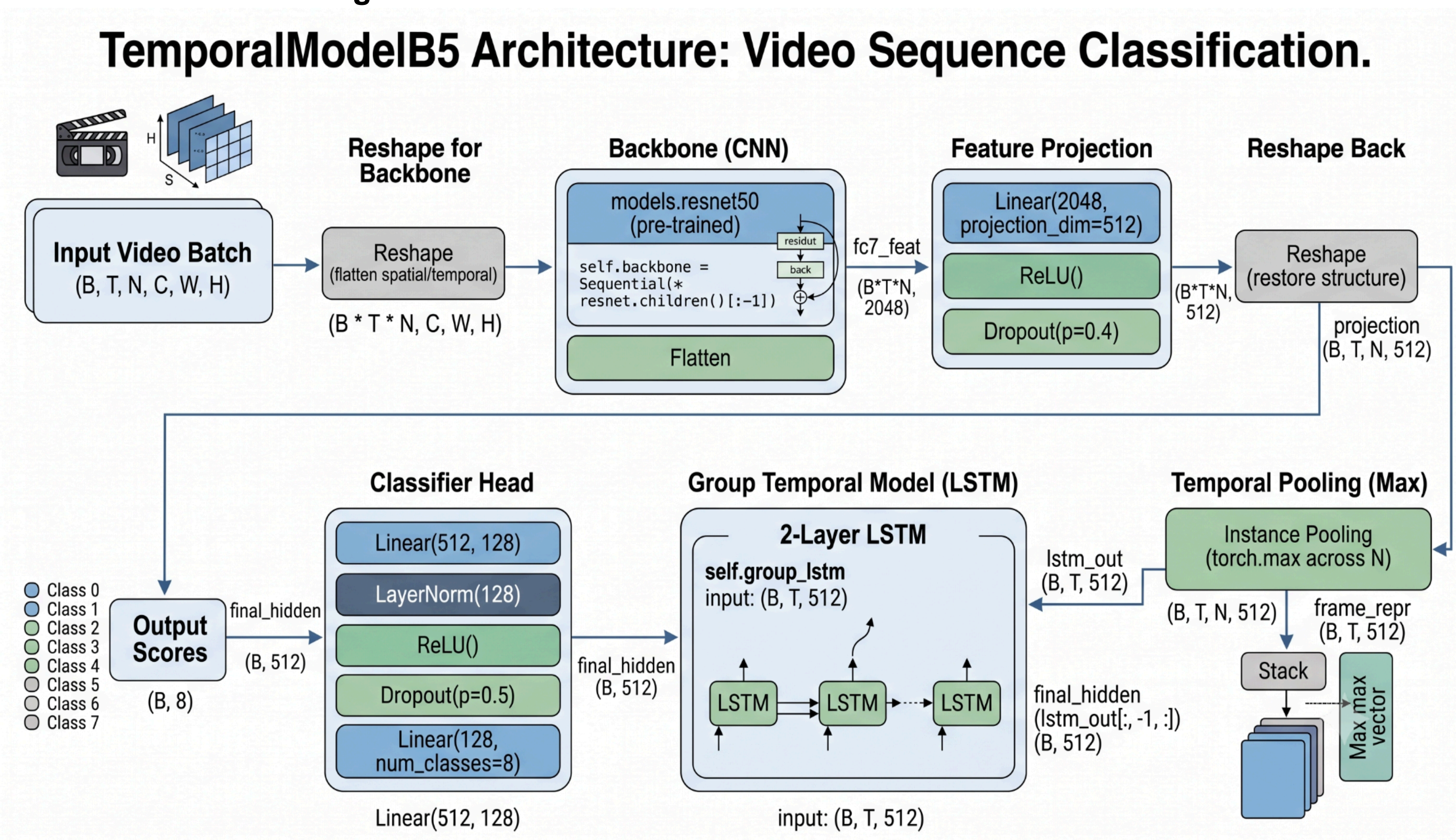
- **Exp Four**(baseline 4: Image classifier using LSTM); I used(learning rate = 1e-4, weight_decay = 1e-4, Epochs = 10, Batch_size = 4)




```
Epoch [1/10] -> Loss: 1.4232 | Training Accuracy: 43.46%
Epoch [2/10] -> Loss: 0.7288 | Training Accuracy: 73.60%
Epoch [3/10] -> Loss: 0.5246 | Training Accuracy: 81.67%
Epoch [4/10] -> Loss: 0.4143 | Training Accuracy: 86.14%
Epoch [5/10] -> Loss: 0.3567 | Training Accuracy: 88.29%
Epoch [6/10] -> Loss: 0.2713 | Training Accuracy: 91.20%
Epoch [7/10] -> Loss: 0.2435 | Training Accuracy: 91.81%
Epoch [8/10] -> Loss: 0.1751 | Training Accuracy: 94.39%
Epoch [9/10] -> Loss: 0.1253 | Training Accuracy: 95.98%
Epoch [10/10] -> Loss: 0.1210 | Training Accuracy: 96.08%

🔬 Evaluating Model Performance Against Unseen Test Set...
📊 Final Accuracy on the Test Set: 73.83%
```

- **EXP Five**(baseline 5: Taking the fc7 features for all players, then max pool over players to get representation for whole image, pass to LSTM2(group LSTM) to keep temporal features, then classify over 8 classes)
- ***Achieving 92.65% on testset***

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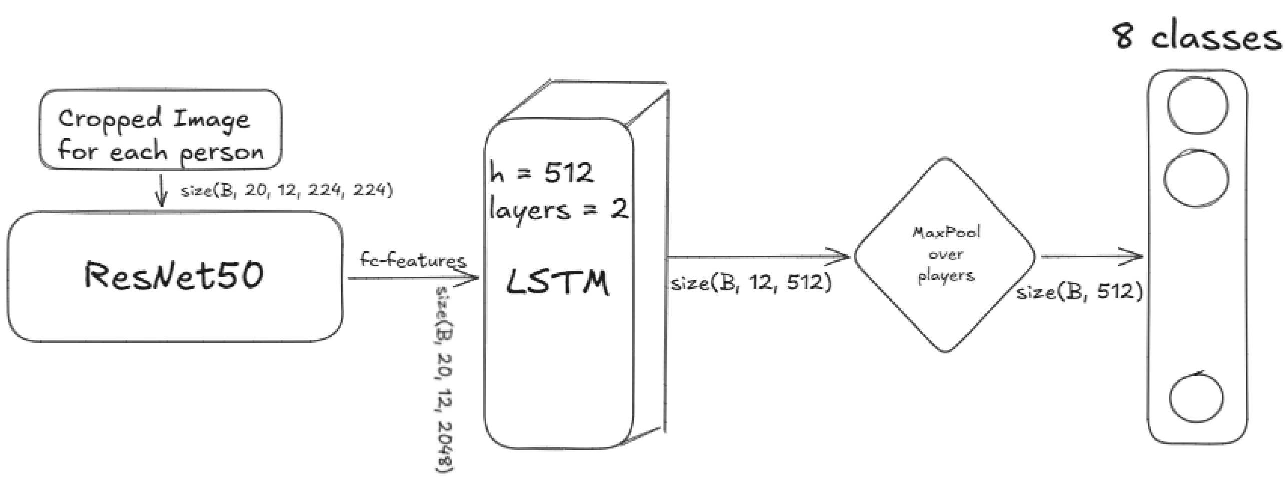
- **EXP Six**(baseline 6: Cropped players over sequence to resnet50, then takes the fc7 features to LSTM1(on player) to detect temporal actions for each player, then max pool over player to get one good representation, then classify into 8 classes); I used (learning rate = 1e-4, weight_decay = 1e-3, Epochs = 20, Batch_size = 4)
- ***Achieving 91% on test set***



**Final Exp.**(Full model: As the previous model, but another LSTM2 added to keep temporal action on the whole image)

***Achieving 93.9% on testset***